

RARITONE DATABASE SCHEMA DOCUMENTATION

Project Name: Raritone Virtual Try-On Platform

Module: Database Design & Schema Documentation

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1. Introduction

The Raritone platform is an AI-powered virtual try-on system that enables users to upload their avatars, store body measurements, browse products, and visualize clothing on their avatars using artificial intelligence. The database is designed using MongoDB and Mongoose to ensure scalability, flexibility, and efficient relationship management between users, products, avatars, and AI-generated outputs.

The database architecture consists of six major schemas:

1. User Schema
2. Product Schema
3. Measurement Schema
4. TryOn Schema
5. Avatar Schema
6. ProductMapping Schema

These schemas work together to provide seamless user management, product management, virtual try-on functionality, and AI-based size recommendations.

2. User Schema

Purpose

The User Schema manages user authentication, authorization, profile information, and account-related activities.

Main Fields

Field Name	Type	Description
_id	ObjectId	Unique user identifier
name	String	User full name
email	String	User email address
password	String	Encrypted password
role	String	User role (Admin/User)
phone	String	Contact number
avatar	ObjectId	Avatar reference
createdAt	Date	Account creation time
updatedAt	Date	Last update time

Relationships

- One User can have one Avatar.
- One User can have multiple Measurements.
- One User can perform multiple Try-On operations.

AI Usage

- Personalization of recommendations.
- User-specific fitting predictions.
- History-based recommendation generation.

3. Product Schema

Purpose

The Product Schema stores product catalog information used throughout the application.

Main Fields

Field Name	Type	Description
id	ObjectId	Product identifier
name	String	Product name
description	String	Product description
category	String	Product category
price	Number	Product price
stock	Number	Available quantity
images	Array	Product images
sizeOptions	Array	Available sizes
createdAt	Date	Product creation date
updatedAt	Date	Last update date

Relationships

- One Product can be used in multiple Try-On sessions.
- One Product can have multiple ProductMappings.

AI Usage

- Product images are processed by virtual try-on models.
- Product metadata supports recommendation systems.
- Size information assists fitting algorithms.

4. Measurement Schema

Purpose

The Measurement Schema stores body measurements required for size prediction and virtual fitting.

Main Fields

Field Name	Type	Description
_id	ObjectId	Measurement identifier
userId	ObjectId	User reference
chest	Number	Chest measurement
waist	Number	Waist measurement
hip	Number	Hip measurement
shoulder	Number	Shoulder measurement
height	Number	User height
weight	Number	User weight
createdAt	Date	Record creation time

Relationships

- Multiple Measurement records belong to one User.

AI Usage

- AI-based size recommendation.
- Garment fitting prediction.
- Personalized try-on accuracy improvement.

5. Avatar Schema

Purpose

The Avatar Schema stores user avatar information used for virtual try-on generation.

Main Fields

Field Name	Type	Description
id	ObjectId	Avatar identifier
userId	ObjectId	User reference
avatarImage	String	Avatar image URL
gender	String	User gender
bodyType	String	Body type classification
createdAt	Date	Upload timestamp

Relationships

- One Avatar belongs to one User.
- One Avatar can be associated with multiple Try-On records.

AI Usage

- Avatar serves as the base image for virtual try-on.
- Computer vision detects body structure.
- AI overlays garments onto avatar images.

6. TryOn Schema

Purpose

The TryOn Schema stores AI-generated virtual try-on results and processing information.

Main Fields

Field Name	Type	Description
_id	ObjectId	Try-On identifier
userId	ObjectId	User reference
productId	ObjectId	Product reference
avatarId	ObjectId	Avatar reference
outputImage	String	Generated result image
status	String	Processing status
createdAt	Date	Try-On timestamp

Relationships

- Many Try-On records belong to one User.
- Many Try-On records reference one Product.
- Many Try-On records reference one Avatar.

AI Usage

- Stores generated virtual fitting results.
- Maintains AI prediction history.
- Supports future model evaluation and improvements.

7. ProductMapping Schema

Purpose

The ProductMapping Schema stores AI-generated mapping data between products and avatars.

Main Fields

Field Name	Type	Description
_id	ObjectId	Mapping identifier
productId	ObjectId	Product reference
avatarId	ObjectId	Avatar reference
fitScore	Number	AI fit confidence score
recommendedSize	String	Suggested size
createdAt	Date	Mapping timestamp

Relationships

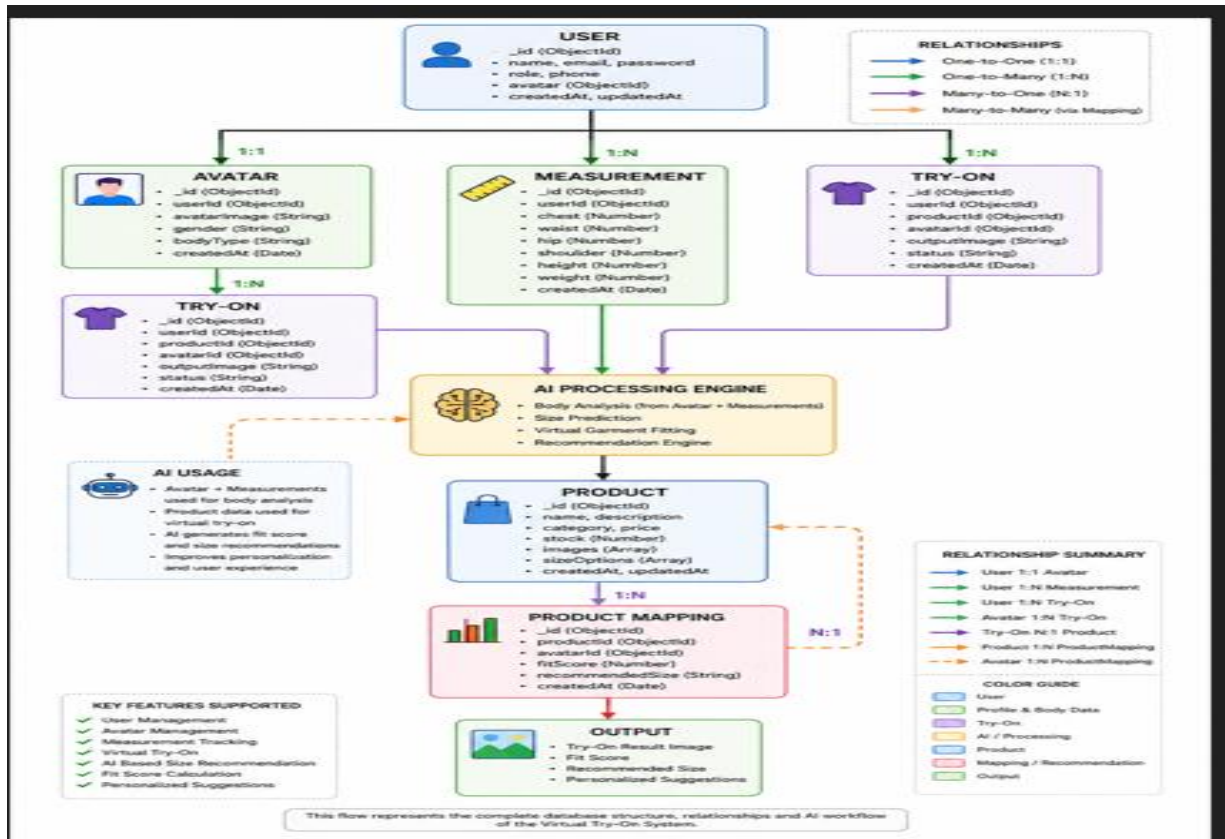
- Multiple mappings belong to one Product.
- Multiple mappings belong to one Avatar.

AI Usage

- Stores AI-generated fitting results.
- Generates personalized size recommendations.
- Improves recommendation engine accuracy.

8. Database Design Architecture

Entity Relationship Diagram



Design Principles

Normalization

The database structure minimizes redundancy by separating users, products, avatars, and measurements into dedicated collections.

Scalability

Each schema is independently scalable, allowing future expansion without affecting existing modules.

Flexibility

MongoDB's document-based architecture enables easy addition of new AI-related fields and metadata.

Performance

References between collections reduce duplication and improve query efficiency.

9. Schema Verification Report

User Schema Verification

- ✓ User registration successful
- ✓ Login authentication working
- ✓ Role-based authorization supported
- ✓ Avatar linkage verified

Product Schema Verification

- ✓ Product creation successful
- ✓ Product retrieval functioning
- ✓ Image storage supported
- ✓ Inventory tracking available

Measurement Schema Verification

- ✓ Measurement data stored correctly
- ✓ User associations validated
- ✓ Data accessible for AI processing

Avatar Schema Verification

- ✓ Avatar upload functioning
- ✓ Avatar retrieval successful
- ✓ User-avatar relationship verified

TryOn Schema Verification

- ✓ User-product-avatar mapping validated
- ✓ Output image storage verified
- ✓ Try-On history maintained

ProductMapping Schema Verification

- ✓ Product-avatar relationships stored correctly
- ✓ Fit score generation supported
- ✓ Size recommendation storage verified

10. Conclusion

The Raritone database architecture provides a robust foundation for virtual try-on operations and AI-powered recommendations. The schema design successfully supports user management, product management, avatar handling, measurement tracking, try-on generation, and intelligent size prediction.

Key Benefits:

- Modular architecture
- Efficient relationship management
- AI-ready data structure
- Scalable MongoDB implementation
- Support for future recommendation enhancements

The database design is production-ready and capable of supporting large-scale virtual try-on operations with AI integration.